Chan Lok Tung, Selene

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Education

National University of Singapore (NUS) | *Scholar of Science & Technology Undergraduate Scholarship* Aug 2025 - Present

Graduating in July 2029, Bachelor of Computing in Artificial Intelligence, Minors in Data Analytics & Language Science (GPA: 4.3/5.0)

- Coursework: Machine Learning optimization, database systems, software engineering, probability & statistics

Stanford University | *Achievement of Completion in International Honors Program 2026* (GPA: 4.0/4.0) Jun - Aug 2026

- Coursework: modern research methods in Cognitive Science, design & analysis of algorithms
- Research proposal: *Savings in Relearning After Machine Unlearning*: whether Language Models relearn erased knowledge faster than when they learn new knowledge

Skills

- Machine Learning: neural networks, optimization, model benchmarking, multi-criteria decision analysis
- Programming Languages: Python, JavaScript, SQL
- Frameworks and tools: Git, Flask, React, PostgreSQL

Key Projects

Sustainable Aviation Fuel (SAF) Supply Chain Optimization Model, AutonomousSG 2026 Airbus Challenge, Singapore Oct 2026

- Developed a Mixed-Integer Linear Programming (MILP) model using Python, PuLP, HiGHS to optimize depot and refinery locations and biomass-to-SAF transportation across Kalimantan, Indonesia
- Built an interactive Streamlit dashboard visualizing facility locations, material flows, cost breakdowns, and operational decisions

Autonomous Machine Learning Research Agent for Recommender Systems, TikTok TechJam 2026, Singapore Aug 2026

- Built an AI agent that iterates the full ML research loop: forming hypotheses, patching its own code, executing iterative optimization loops, verifying the results, then forming hypotheses again
- Evaluated its performance against the Factorization Machine (FM) baseline on the KuaiRand-Pure short-video ranking benchmark

Ensemble Model for Clinical Decision in Women's Hormonal Health, Hack Nation 6th Global AI Hackathon, US Jul 2026

- Built a three-model ensemble, with male-only, female-only, mixed data, on ~3,700 patient records for PCOS and thyroid decisions
- Shipped a Flask app with separate doctor and patient views for report upload, AI summaries, and centralized records across hospitals
- Implemented doctor alerts when patient-reported symptoms cannot be confidently assessed, and when information was insufficient

Social Network Simulation of Misinformation Spread, NUS Orbital, Singapore | ***Second-Tier Achievement*** May - Aug 2026

- Modeled misinformation spread with an Independent Cascade Model over a directed graph, with activation probability driven by virality of information, weights assigned to user actions, and node susceptibility
- Assessed users' information-evaluation behavior within the simulation using a Multi-Criteria Decision Analysis framework, based on decision accuracy, misinformation sharing, and habits of verifying new information
- Built a full-stack application using React, React Router v7, PostgreSQL, and Cytoscape.js

Internship Experience

Research Intern, Qiming Venture Partners, Hong Kong SAR Jul - Aug 2025

- Identified the Middle East as aerospace AI's fastest-growing market, with a forecast compound annual growth rate of 9.31%
- Reviewed published comparisons of Neural Network versus Base of Aircraft Data models for aircraft arrival-time prediction (66.3% more accurate, 96.9% confidence in prediction bounds)
- Examined risks associated with applying digital twins and User-Centric Explainable AI to air traffic control and management
- Profiled 2 air traffic control vendors, Traffic Control Technology Company Limited (688015.SH) and NoamAI (startup)

Research Intern, Unicorn Partners, Hong Kong SAR Jul 2025

- Analyzed the latest AI products launched at Google I/O 2025 and potential implications for stakeholders' adoption
- Examined how AI agent templates enable non-technical industry experts to build their own AI solutions to domain-specific problems
- Argued the role of Retrieval Augmented Generation (RAG) in personalizing AI agents

Leadership Experience

Deputy Director, Business Committee, NUS Students' Union (NUSSU), Singapore Oct 2026 - Present

- Leading corporate sponsors' partnerships by developing sponsorship proposals that support NUSSU initiatives

- Executing corporate events and sponsorship activities, overseeing budgets, logistics, and promotional materials
- Managing and updating the sponsor database, monitoring partnership performance, refining sponsorship guidelines and processes to improve efficiency and compliance

Vice President, Outgoing Global Exchange, AIESEC in NUS, Singapore Jan 2026 - Present

- Led a 16-person committee helping students access overseas volunteering and internship opportunities in 15 countries
- Raised students' participation rate in overseas exchange programmes from 19.2% to 23.8% through targeted campaigns

Additional Information

Languages: English, Cantonese, Mandarin (native proficiency)